

Q1.

Arteries carry blood away from the heart. The largest artery is the aorta, which carries oxygenated blood to the whole body.

Q3.

Chloroplasts contain the green pigment chlorophyll a and chlorophyll b, which absorb light energy. The two stages are the light reaction and the dark reaction.

Q2.

The chloroplast is the organelle primarily involved in photosynthesis.

Q5.

The alveolus is a small air sac surrounded by a network of blood capillaries. Oxygen diffuses from the alveolar air space into the capillary and carbon dioxide diffuses out in the opposite direction.

Q7.

The nephron begins with the Bowman's capsule which encloses the glomerulus. Filtrate then passes into the proximal tubule, then the loop of Henle, and finally into the collecting duct.

Q6.

The human digestive system starts at the mouth and continues through the oesophagus into the stomach, where food is churned with gastric juice. The liver secretes bile and the pancreas secretes pancreatic

juice, both of which empty into the small intestine.

Most of the absorption of digested food takes place in the small intestine, mainly in the ileum, because it has villi that increase the surface area.

Q9.

Transpiration is the loss of water vapour from the aerial parts of a plant, mostly through the stomata of leaves.

Two factors that increase the rate are high temperature and increased wind speed.

Q8.

Palisade mesophyll cells are long and column shaped and are packed tightly with chloroplasts, so they absorb the most light. Spongy mesophyll cells are loosely packed with large air spaces between them for gas exchange.

Q11(b).

Move the pot to a south facing window so that it gets more sunlight every day.

Q12.

Minute ventilation = tidal volume x breathing rate
= 0.5×12
= 6 L per minute.

Q13.

Alveolar ventilation = (tidal volume - dead space) x rate
= $(0.5 - 0.15) \times 12$
= 0.35×12
= 4.2 L per minute.

Q15.

The mitochondria is the powerhouse of the cell and carries out aerobic respiration.